

Kirtan Gangani

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👤 KirtanGangani

Research Areas

Vision-Language Models, Multimodal Benchmarking, Thermal Computer Vision, AI for Sustainability

Education

Ajeenkya DY Patil University (ADYPU)

Sept 2021 – Jun 2025

B.Tech in Computer Engineering (GPA: 9.41/10.00)

Pune, India

- Minor: Mechatronics and Robotics Engineering
- Thesis: Sentiment Analysis of Indian Political Tweets Using Ensemble of Transformer Models and LLMs
- Relevant Coursework: Digital Image and Video Processing, Data Mining and Machine Learning, Artificial Intelligence, Linear Algebra and Differential Calculus, Natural Language Processing

Publications

- [1] A. Shrivastava, **K. Gangani**, L. Jain, M. Goel, and N. Batra. “*ThermEval: A Structured Benchmark for Evaluation of Vision-Language Models on Thermal Imagery.*” In **KDD 2026**.

Research Experience

Research Associate

Aug 2025 – Jul 2026

Sustainability Lab, Indian Institute of Technology Gandhinagar (IITGN)

Supervised by: Prof. Nipun Batra

- Co-developed ThermEval, a structured benchmark of 70,000+ thermal VQA pairs across 7 tasks, designed to evaluate foundational primitives required for thermal vision-language understanding.
- Evaluated 25 open- and closed-source VLMs spanning 0.3B to 200B+ parameters, uncovering three systematic failure modes: inability to perform temperature-grounded reasoning, sensitivity to colormap transformations, and over-reliance on language priors.
- Contributed to ThermEval-D, the first dataset providing dense per-pixel temperature maps with semantic body-part annotations across diverse indoor and outdoor environments.
- Conducted follow-up research on thermally-grounded image captioning for VLMs, developing methods that infuse physical thermal context to improve caption quality over RGB-adapted baselines.

Research Intern

May 2025 – Jul 2025

Sustainability Lab, Indian Institute of Technology Gandhinagar (IITGN)

Supervised by: Prof. Nipun Batra

- Contributed to ApneaEye, an AI/ML initiative developing a cost-effective, thermal-video-based diagnostic tool for non-invasive sleep apnea detection, significantly reducing reliance on specialized lab equipment.
- Designed and implemented an image classification model to identify patient positions from thermal video footage, a critical preprocessing step in the detection pipeline.
- Performed large-scale data annotation, processing thousands of thermal images to build a robust training dataset for the machine learning pipeline.
- Developed a web-based interface for ApneaEye enabling seamless tracking and visualization of patient data, including longitudinal graphs, via unique patient IDs.

Awards & Honors

- Bronze Medal, awarded for ranking 3rd in the graduating B.Tech. Computer Engineering batch, ADYPU (2025)

Skills

Programming Languages: Python, C++, L^AT_EX

ML / DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace, OpenCV

Data & Viz: NumPy, Pandas, Matplotlib, Seaborn, SciPy, Power BI

Databases: MySQL, PostgreSQL

Tools: Git, Linux, Jupyter

Languages: English (Professional), Gujarati (Native), Hindi (Fluent)